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Romeo And Juliet Meeting

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1 Romeo and Juliet Meeting Problem

Romeo and Juliet agree to meet at a specific time. However, both of them may arrive late. Each person has a delay between 0 and 1 hour. The first person to arrive waits for 15 minutes and leaves if the other person has not arrived.

The aim is to calculate the probability that Romeo and Juliet meet.

2 Theoretical Mathematics

Let

$$X = \text{Romeo's delay in hours}$$

and

$$Y = \text{Juliet's delay in hours.}$$

Since each person can arrive with a delay between 0 and 1 hour, we assume that

$$X \sim U(0,1)$$

and

$$Y \sim U(0,1).$$

Also, the problem states that all pairs of delays are equally likely. Therefore, X and Y are independent random variables.

Hence, the pair (X,Y) is uniformly distributed over the unit square

$$[0,1] \times [0,1].$$

This means that every possible outcome corresponds to one point inside the square

$$0 \leq X \leq 1, \qquad 0 \leq Y \leq 1.$$

The total area of this square is

$$A_{\text{total}} = 1 \cdot 1 = 1.$$

Because the distribution is uniform, probability is equal to area. Therefore,

$$P(\text{event}) = \frac{\text{area of favourable region}}{\text{area of total region}}.$$

Since the total area is 1, the probability is simply equal to the area of the favourable region.

3 Meeting Condition

The first person who arrives waits for 15 minutes. Since

$$15 \text{ minutes} = \frac{15}{60} = \frac{1}{4} \text{ hour},$$

we define the acceptable waiting time as

$$a = \frac{1}{4}.$$

Romeo and Juliet meet if the difference between their arrival times is less than or equal to a. Therefore, the meeting condition is

$$|X - Y| \leq a.$$

For this problem, this becomes

$$|X - Y| \leq \frac{1}{4}.$$

The line

$$Y = X$$

represents the case where Romeo and Juliet arrive at exactly the same time.

The condition

$$|X - Y| \leq a$$

represents all points inside a diagonal band around the line $Y = X$. These are the points where the difference between Romeo's delay and Juliet's delay is not larger than the waiting time.

4 Calculation of the Exact Probability

Instead of calculating the meeting region directly, it is easier to calculate the region where Romeo and Juliet do not meet.

They do not meet if

$$|X - Y| > a.$$

This can happen in two cases:

$$Y > X + a,$$

or

$$Y < X - a.$$

These two inequalities describe two triangular regions inside the unit square.

First, consider the upper triangle:

$$Y > X + a.$$

The boundary of this region is the line

$$Y = X + a.$$

This line intersects the left side of the unit square when $X = 0$. Therefore,

$$Y = 0 + a = a.$$

So one intersection point is

$$(0, a).$$

The same line intersects the top side of the unit square when $Y = 1$. Therefore,

$$1 = X + a.$$

Solving for X, we get

$$X = 1 - a.$$

So the second intersection point is

$$(1 - a, 1).$$

Therefore, the upper non-meeting region is a right triangle with side length

$$1 - a.$$

The area of this triangle is

$$A_1 = \frac{1}{2}(1 - a)(1 - a).$$

Thus,

$$A_1 = \frac{1}{2}(1 - a)^2.$$

There is an identical triangle below the line

$$Y = X - a.$$

Therefore, the area of the lower non-meeting triangle is also

$$A_2 = \frac{1}{2}(1 - a)^2.$$

The total area where Romeo and Juliet do not meet is

$$A_{\text{not meet}} = A_1 + A_2.$$

Substituting the two triangle areas gives

$$A_{\text{not meet}} = \frac{1}{2}(1 - a)^2 + \frac{1}{2}(1 - a)^2.$$

Therefore,

$$A_{\text{not meet}} = (1 - a)^2.$$

Since the total area is 1, the probability that Romeo and Juliet do not meet is

$$P(\text{not meet}) = (1-a)^2.$$

Therefore, the probability that they meet is

$$P(\text{meet}) = 1 - P(\text{not meet}).$$

Hence,

$$P(\text{meet}) = 1 - (1-a)^2.$$

Expanding the expression,

$$P(\text{meet}) = 1 - (1 - 2a + a^2).$$

Therefore,

$$P(\text{meet}) = 1 - 1 + 2a - a^2.$$

So,

$$P(\text{meet}) = 2a - a^2.$$

For this problem,

$$a = \frac{1}{4}.$$

Thus,

$$P(\text{meet}) = 2\left(\frac{1}{4}\right) - \left(\frac{1}{4}\right)^2.$$

This gives

$$P(\text{meet}) = \frac{7}{16}.$$

Therefore, the exact probability that Romeo and Juliet meet is

$$\boxed{P(\text{meet}) = \frac{7}{16}}.$$

As a decimal,

$$\frac{7}{16} = 0.4375.$$

Therefore,

$$\boxed{P(\text{meet}) = 0.4375}$$

or

$$\boxed{P(\text{meet}) = 43.75\%}.$$

5 Monte Carlo Simulation

The exact probability can also be checked using a Monte Carlo simulation.

In the simulation, we generate many random values for Romeo's delay and Juliet's delay. Each delay is a random number between 0 and 1, representing a delay in hours.

For each trial, we generate

$$X_i \in [0, 1]$$

and

$$Y_i \in [0, 1].$$

They meet if

$$|X_i - Y_i| \leq \frac{1}{4}.$$

If this condition is true, we count the trial as a successful meeting.

After N simulations, the estimated probability is

$$\hat{P}(\text{meet}) = \frac{\text{number of successful meetings}}{\text{total number of simulations}}.$$

So,

$$\hat{P}(\text{meet}) = \frac{M}{N},$$

where M is the number of times Romeo and Juliet meet, and N is the total number of trials.

As N becomes larger, the simulated probability should become closer to the exact probability. This follows from the Law of Large Numbers.

6 Comparison Between the Code and the Theory

The theoretical condition for Romeo and Juliet to meet is

$$|X - Y| \leq a.$$

In the code, this is implemented as

```
if abs(romeo_delay - juliet_delay) <= acceptable_delay:
```

This corresponds exactly to the mathematical condition

$$|X - Y| \leq \frac{1}{4}.$$

The theoretical exact probability is

$$P(\text{meet}) = 1 - (1 - a)^2.$$

In the code, this is implemented as

```
def calculate_probability(acceptable_delay):
    return 1 - (1 - acceptable_delay) ** 2
```

Therefore, the theoretical formula and the code formula are the same.

For the specific value

$$a = \frac{1}{4},$$

the code calculates

$$P(\text{meet}) = 1 - \left(1 - \frac{1}{4}\right)^2.$$

This gives

$$P(\text{meet}) = 1 - \left(\frac{3}{4}\right)^2.$$

Therefore,

$$P(\text{meet}) = 1 - \frac{9}{16}.$$

Thus,

$$P(\text{meet}) = \frac{7}{16} = 0.4375.$$

So the exact result produced by the code should be

$$0.4375.$$

The simulation result will usually be close to

$$0.4375,$$

but not exactly equal to it, because it is based on random trials.

7 Results

The estimated probability gets closer to the exact probability.This happens because of the Law of Large Numbers, which states that as the number of independent trials increases, the simulated average approaches the theoretical expected value.

Therefore, the first plot demonstrates that

$$\hat{P}(\text{meet}) \to P(\text{meet})$$

as

$$N \to \infty.$$

In this problem, that means

$$\hat{P}(\text{meet}) \to \frac{7}{16}.$$

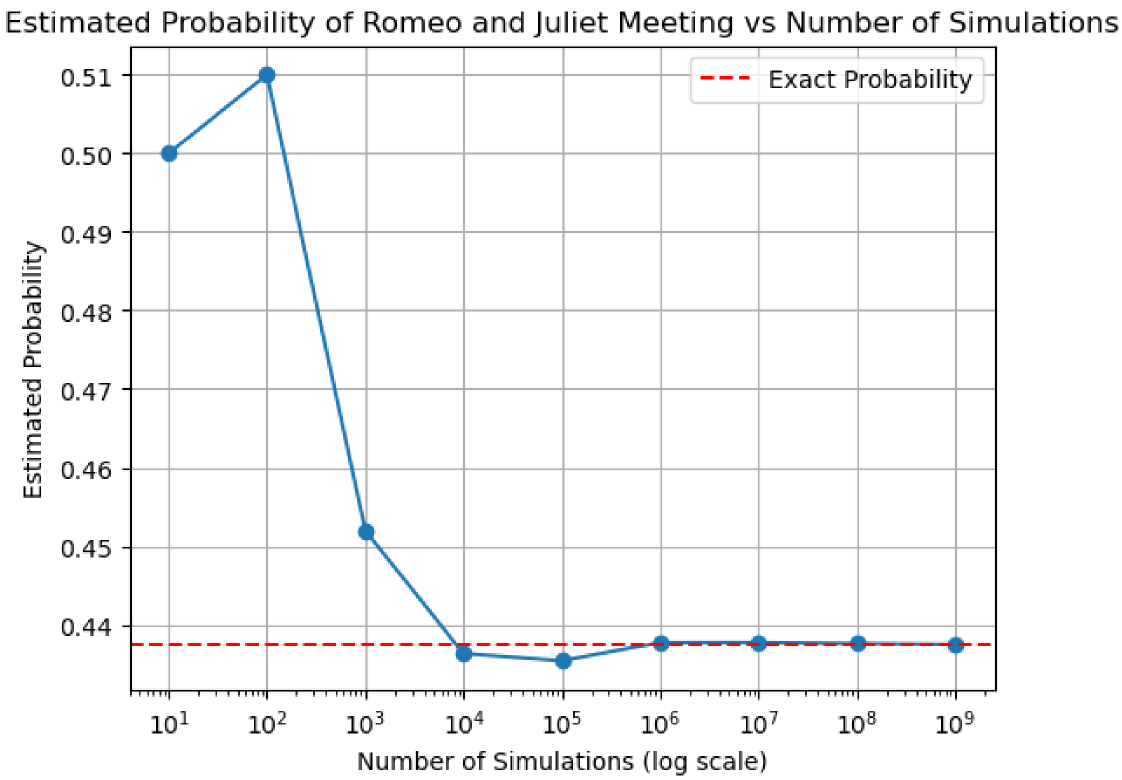

Figure 1: Estimation of probability for different number of trails. It is clear with the more trials the predicted probability reaches the exact probability.

The theoretical probability for a general acceptable delay a is

$$P(\text{meet}) = 1 - (1 - a)^2.$$

Equivalently,

$$P(\text{meet}) = 2a - a^2.$$

Therefore, the exact probability depends on the acceptable delay a.
When

$$a = 0,$$

Romeo and Juliet meet only if they arrive at exactly the same time. In a continuous probability model, the probability of this is

$$P(\text{meet}) = 0.$$

When

$$a = 1,$$

Romeo and Juliet always meet, because both delays are between 0 and 1 hour. Therefore,

$$P(\text{meet}) = 1.$$

For values between 0 and 1, the probability follows the quadratic curve

$$P(a) = 2a - a^2.$$

For example, if

$$a = 0.25,$$

then

$$P(a) = 2(0.25) - (0.25)^2.$$

Therefore,

$$P(a) = 0.5 - 0.0625.$$

So,

$$P(a) = 0.4375.$$

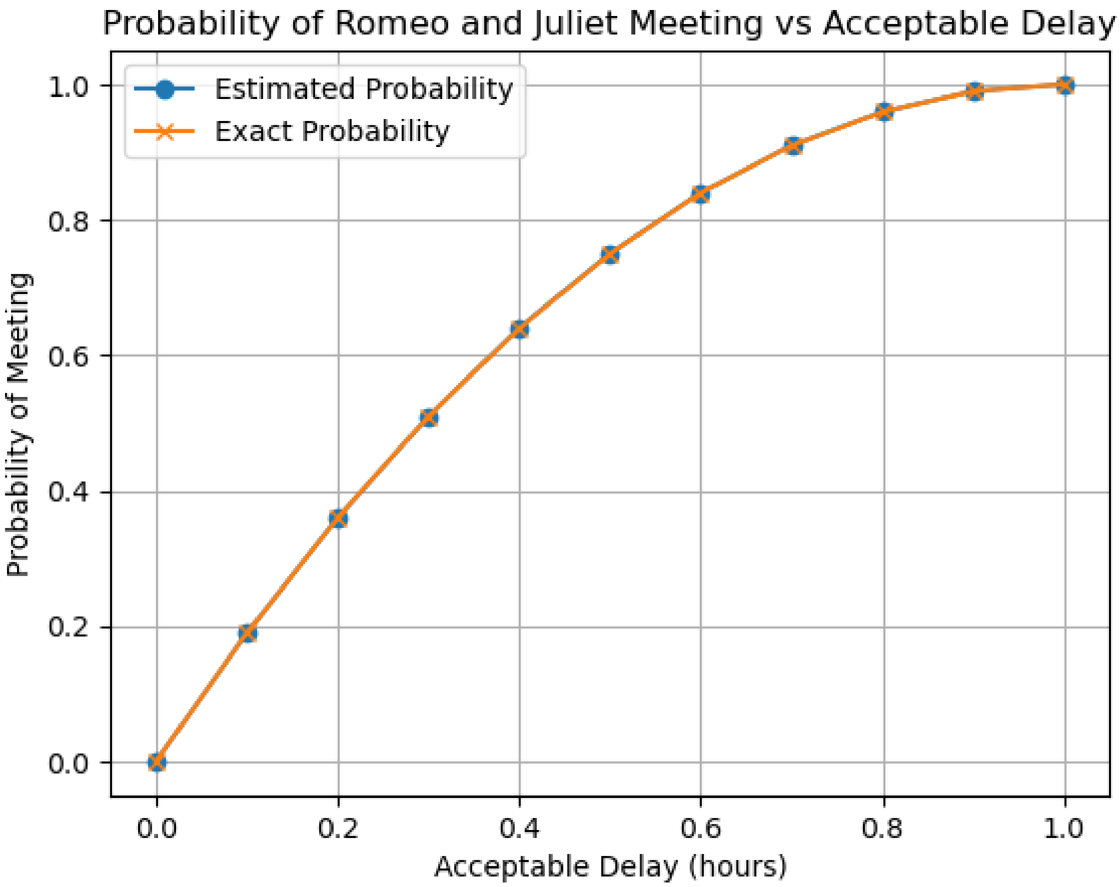

Figure 2: Estimation of probability for different acceptable delays. It is clear with more acceptable delay the probability increases.

The Fig: 2 compares the simulated probabilities with the exact probabilities for different values of a. The two curves should be very close because the simulation uses 10^6 trials.

8 Final Conclusion

- The theoretical probability that Romeo and Juliet meet is 43.75%.
- The Monte Carlo simulation confirms the theoretical result, because the estimated probability becomes closer to the exact probability as the number of simulations increases.
- The small difference between the exact result and the simulated result is caused by randomness in the simulation.
- Both approaches use the same main simulation idea: Romeo's and Juliet's delays are randomly generated between 0 and 1, and the program checks whether their difference is less than or equal to the acceptable delay.
- Both approaches use the same exact probability formula, so the final theoretical result is the same.
- My original code includes two random variables outside the simulation function, but these variables are not used in the final calculation.
- Overall, there is no difference in the mathematical approach or final result; the differences are mainly related to code clarity, readability, and organization.

A Python Code

All code used to generate the results, figures, and simulations in this report is publicly available at:

Project GitHub Repository